

Review:

Thanks to the authors for this contribution. The idea presented is relatively new to the dInSAR technique for dSWE and it presents interesting discussion on multilooking window selection for processing. However, I have found issues that prevent me from recommending its publication without a major revision. The paper needs to broaden theoretical discussion, including selected references and provide a more thoughtful discussion on their findings.

General Comments:

There are important papers that should be consulted as they discuss coherence for non-stationary processes. Touzi et al., 1999 models differently the non-stationarity. Yueh et al., 2026 directly addresses the topic presented here. The latter is particularly important here. Also consider Rott et al., 2003, although the formulation is for sub-pixel scale.

An important addition to the paper would be to quantify both the absolute value and the argument of M with respect to standard deviation or relevant metrics within the window. Now, the reader cannot quantify the any impact of the variability. It is necessary to better quantify the magnitude of the effect. Consider the following:

1. Choose a window size and generate phase samples (“per-cell deviation from mean”) drawn from a distribution. For example, Gaussian $N(0, \sigma)$ or consider other distribution that captures the spatial distribution within the window.
2. Calculate complex M , repeat enough times till convergence.
3. Repeat for different values of sigma. This would provide an idea of the effect of M for different window variabilities, instead of the arbitrary examples presented in Figure 1.

This can also be calculated analytically but of course it’s more tedious. Consider this both in absolute dSWE and radians. Consider different distributions. For example, in case the underlying distribution is Gaussian of δ , or a symmetric distribution in general, I would expect that $E\{\arg(M)\} = 0$. This is your case in Figure 1 (middle). Then, the estimated phase from the window will not be skewed. The reader needs to understand already in section 2 what causes the skewing.

How does the typical multilooking window look like in terms of ΔSWE ? Which distribution approximate it better? This could be linked to your dataset. Based on your results (Figure 5), the distribution of ΔSWE may not be symmetrical. While this is partially addressed in Section 4.2, I think the aggregation over the whole dataset does not allow understanding of the individual multilook variability.

Another important consideration are variations of local incidence angle in the multilooking window. Although the sensitivity is smaller than ΔSWE , changes in LIA within the multilooking window will produce the same effect even for constant ΔSWE . In particular to your dataset, the northeast area seems to have lower values of M, but also presents more topographical variations.

Can you report the spatial baseline between NISAR acquisitions? The orbital tube for NISAR is small, but I wonder if small spatial baseline between passes can impact the coherence in some areas of your images.

A distinction between posting, resolution and multilook is needed. The multilooked domain, either 5x6 or 13x16, does not correspond to 20 or 81 ground meters, as multilooking is calculated in radar coordinates. Range resolution also depends with incidence angle. Comparing NISAR coherence and Lidar-derived M needs to be done with care (e.g., Section 4.3).

The terminology “within-window”, “sub-window”, and “footprint” is used somewhat interchangeably throughout the manuscript.

Specific Comments:

The abstract and introduction does not stress enough the main problem. Phase is highly sensitive to ΔSWE and ΔSWE is highly spatially variable. Also high decorrelation makes large multilooking an attractive option to deal with noise at a cost of spatial resolution.

Section 2:

Somewhere along the first paragraph it should be commented that this “averaging” is performed spatially.

Equation 2 should be introduced more clearly.

Figure 1 requires more explanation, describe how many pixels are you using, the central frequency you are using (as later described this is critical). Also consider discussing if the presented dynamic range is reasonable for SAR revisit times. Do you observe $\pm 8\text{cm}=80\text{mm}$ ΔSWE often?

I think you need to differentiate between real noise free ΔSWE or $\Delta\phi$ and observed values. For example, in Line 86 it is unclear if δ refers to stochastic noise or ΔSWE deviations from the mean. If the process is stationary, then $\delta = 0$ for every pixel. The section already discusses in terms of estimated phase variance when in my understanding, you are discussing the effect of the underlying distribution of ΔSWE .

Section 3.4:

When using the LIA=40 for the whole image, the topography is ignored, which varies sub-window variability.

Section 3.6:

The claim on the Circular Gaussian as the limitation is unclear.

Equation 8 is based on stationary samples. Since the central discussion of the paper is the non-stationarity I think commenting the validity of this equation is in order.

Section 4.5:

The results, in particular related to C-band are quite pessimistic. How do these compare to literature, for example Oveisgharan et al., 2024, Oveisgharan et al., 2026, Bovenga et al., 2025 among others? This requires reconciliation with previous papers.

Section 5.2:

I don't agree with this. For tower imagery I would not expect high spatial variability within the multilook window. The whole window may be smaller than a typical satellite resolution cell.

Snow depth/SWE for sea ice snow, is typically smaller than land. This is hinted on line 433. This is why I believe you need to quantify either analytically or numerically the effect of phase variance in the complex coherence.

Specific Comments:

Line 22: how "approximately weekly observations" are met? Which constellation? S1? Where?

Line 48 to 50: These two sentences discuss different issues, variability in the pixel and within the estimation window, and are provided without reference.

Line 50: Not completely sure if the original quote refers to your problem or phase unwrapping errors, as differences larger than 2π cannot be recovered with some unwrapping tools.

Line 74: PDF require definition?

Line 86: I would suppose that all terms in this section are already noise-free, then how is δ decorrelation noise alone?

Line 102: This is repeated.

Line 103: not necessarily true for all frequencies.

Line 164: Int.phase is defined between $(-\pi, \pi]$.

Line 328 and 333: Again the effect of the LIA could be affecting the non-stationarity.

Line 342: I don't agree with this, see my points regarding Section 2.

Line 406: I think those paper start my calculating coherence spatially, so I don't think saying they use temporal rather than spatial averaging is right, can you check?

Line 412: drop the formula of the phase closure as neither it or z are defined or used elsewhere.

Figure 7 refers to sub-pixel on the title.